

Name: _____

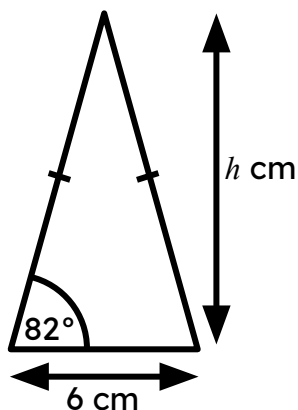


Problem solving with trigonometry

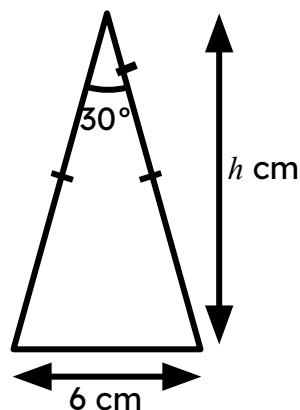
Task A: Calculating areas

1) Calculate the area of each triangle, to 1 decimal place.

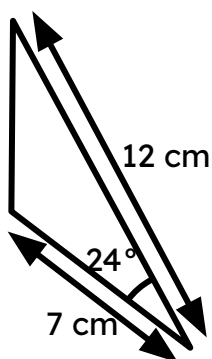
b)



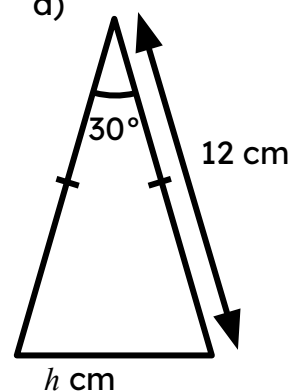
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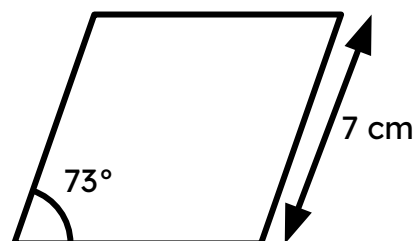
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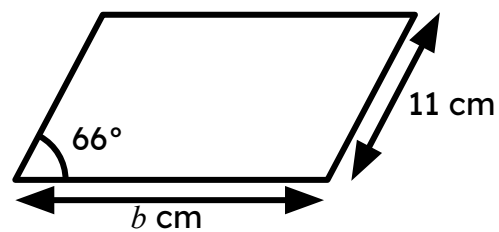
d)



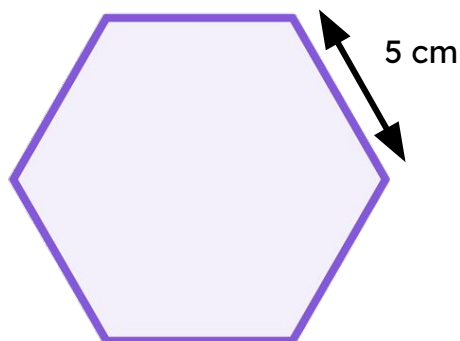
2) Calculate the area of this rhombus, to 1 decimal place.



3) Calculate the area of this parallelogram, to 1 decimal place, given that its perimeter is 54 cm.

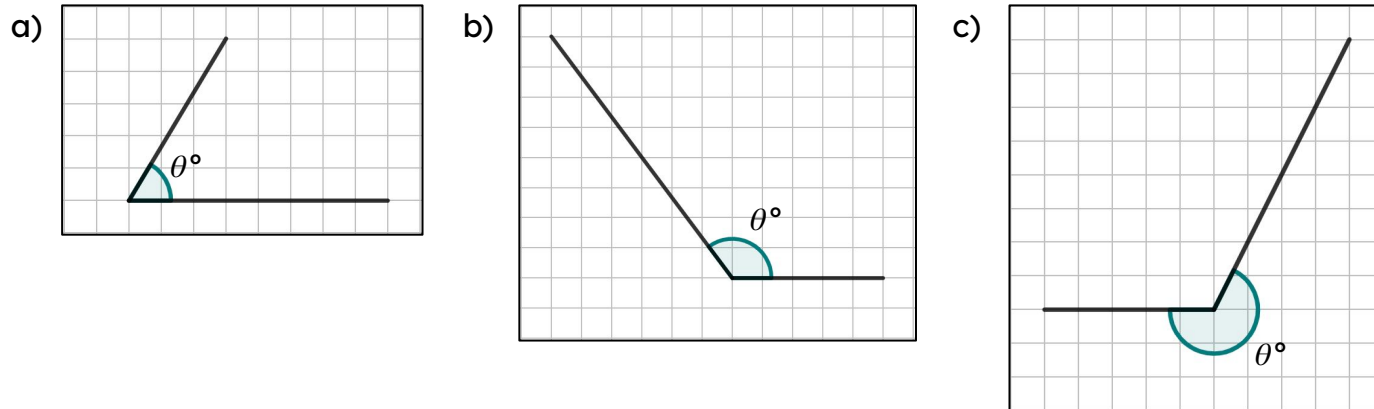


4) Calculate the area of this regular hexagon, to 1 decimal place.



Task B: Calculating angles

1) Calculate the marked angles on each diagram.



2) A yacht is given the instruction to sail 20 km due East and then 60 km due North. If the yacht is already facing East, what angle should it turn to be facing directly towards its final position?

3) A helicopter travels 30 km due North and then rotates θ° , before travelling 50 km in that direction. The helicopter finishes directly due West of the starting point. What angle of turn did they make?